



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

AI-enabled wireless chip

A single-chip wireless device, SYN765x, brings AI processing and connectivity together for edge systems such as appliances, home automation products,



and industrial IoT equipment. The chip combines Wi-Fi 7, Bluetooth LE 6.0, and Thread/Zigbee with on-chip AI compute, enabling control and signal-processing tasks to run locally rather than relying on the main processor. It can operate as a co-processor with a host processor or microcontroller, or run in a host-less mode using its own memory and compute resources, which in some designs removes the need for a microcontroller and lowers system complexity and cost. The device also uses Wi-Fi and Bluetooth signals for sensing tasks such as presence detection, motion tracking, and proximity awareness, using Channel State Information with on-device machine learning for Wi-Fi sensing and Bluetooth channel sounding for distance measurement. By combining wireless connectivity, sensing, and AI processing on a single chip, the device reduces system size and component count while enabling more processing at the network edge.

Synaptics Incorporated
<https://www.synaptics.com>

Applications processor

NXP Semiconductors has launched the i.MX 93W applications processor,

expanding its i.MX 93 family with a platform built for physical AI at the edge. The company claims it is the industry's first application processor to



combine a dedicated AI neural processing unit (NPU) with secure tri-radio wireless connectivity in a single package, letting customers replace up to 60 discrete components. Its dual-core processor and dedicated NPU deliver up to 1.8 eTOPs of AI performance. The i.MX 93W integrates scalable edge compute, AI acceleration, and secure wireless connectivity, supporting a wide range of applications—from industrial gateways and energy monitors to smart homes and healthcare systems.

NXP Semiconductors
<https://www.nxp.com>

System-on-Module

Grinn has introduced the Grinn AstraSOM261x, a SoM based on the Synaptics Astra 2610 processor family. The company says the module is the smallest



SoM in this processor line for edge AI systems. It provides computing and AI functions in a small form factor. The AstraSOM261x is the result of a collaboration between Grinn and Synaptics.

It combines Grinn's hardware design with the Astra AI processor line to bring edge AI technology to developers. The module uses Synaptics' Torq Edge AI platform, which includes the Coral NPU from Google Research with dynamic operator support.

Grinn Global
<https://grinn-global.com>

IoT module

Blue has introduced Notecard for Skylo, a module that combines satellite, cellular, and Wi-Fi connectivity without requiring subscriptions or commitments. It supports automatic



failover across NTN Satellite, powered by Skylo, Narrowband Cellular, and Wi-Fi, and adds cloud connectivity with multi-network failover to

any narrowband device. The module targets industries such as transportation, logistics, energy, and commercial equipment, enabling prototyping and scalable deployment of narrowband solutions at lower cost. For example, a logistics asset can use Wi-Fi in depots, cellular on the road, and satellite in remote areas, with the module automatically switching networks as needed. Notecard for Skylo requires no extra hardware, no satellite subscription, and no minimum usage. Businesses can access satellite failover with pay-as-you-go pricing that scales with use. The module ensures continuous connectivity, with wireless connections falling back to secondary or tertiary networks if the primary fails.

Blues
<https://blues.com>